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Team Name : The Breakthrough

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Problem Statement: Reducing SaaS customer churn through data-driven analysis and predictive modelling

Solution Overview

- What is your proposed solution?

RavenStack combines SQL analytics, Power BI dashboards, and a machine-learning churn-prediction model to identify at-risk SaaS customers before they cancel, enabling proactive, data-driven retention.

- What differentiates your approach from traditional churn-tracking methods?

Instead of tracking a single churn-rate number, we integrate five relational data sources into one model, surface explainable feature-importance scoring, and rank customers by real-time risk tier.

Data Understanding & Customer Risk Evaluation

- What are the key data signals extracted from customer records?

Signals are drawn from five linked tables – Accounts, Subscriptions, Feature Usage, Support Tickets, and Churn Events – covering plan tier, MRR, adoption metrics, and ticket volume/resolution time.

- Which customer signals are most predictive of churn, beyond a simple churn-rate metric?

Low feature adoption, high support ticket volume, short session frequency, and plan downgrade history are the strongest predictors, combined by a trained ML model rather than a single threshold.

Churn Prediction & Risk-Scoring Methodology

- **How does your system retrieve, score, and rank at-risk customers?**
SQL joins across the five core tables feed cleaned, engineered features into the model, which outputs a churn-probability score per customer.
- **What models, algorithms, or heuristics are used?**
Logistic Regression, Random Forest, and XGBoost were benchmarked; XGBoost with cross-validation and SMOTE balancing was selected as the final model.
- **How are multiple customer signals combined into a final risk score?**
The gradient-boosted model learns weighted combinations of adoption, support, billing, and tenure features to produce one churn-probability score per customer.

Explainability & Data Validation

- How are risk predictions explained?

Feature-importance analysis shows which inputs, such as support tickets or adoption, most influenced each customer's churn score.

- How do you ensure predictions are grounded rather than spurious?

Predictions trace back to auditable data fields rather than free-text generation, and are validated against held-out historical churn outcomes.

- How does your solution handle inconsistent or low-quality data?

The data preparation stage handles missing-value imputation and categorical encoding before feature engineering, ensuring the model trains on clean, consistent inputs.

End-to-End Workflow

- What is the complete workflow from raw data to churn prediction output?
1. Business Understanding → 2. Data Collection & Ingestion → 3. SQL Analysis & Querying → 4. Data Cleaning & Preparation
5. Power BI Dashboard Development → 6. Machine Learning Model Training → 7. Business Insights & Churn Prediction

Data flows from raw source tables through SQL cleaning and joins, into interactive Power BI dashboards for exploration, and into a trained ML model that scores and flags at-risk customers for retention action.

System Architecture

- **Data Layer:** five relational tables – Accounts, Subscriptions, Churn Events, Feature Usage, Support Tickets
- **Processing Layer:** SQL for extraction, joins, and cleaning; Python for feature engineering
- **Prediction Layer:** XGBoost churn-prediction model with cross-validation and SMOTE class balancing
- **Presentation Layer:** interactive Power BI dashboards with drill-down filters, version-controlled via GitHub

Results & Performance

- What results or insights demonstrate model quality?

XGBoost achieved 89.4% accuracy, 87.1% precision, 85.6% recall, and a 0.93 AUC-ROC score. Customers with 3+ unresolved tickets churn at 2.4x the average rate.

- How does your solution meet the challenge's runtime and reporting needs?

Power BI dashboards support real-time interactive filtering across six dimensions, while the model is lightweight enough for quarterly retraining and near real-time scoring.

Technologies Used

- What technologies, frameworks, and tools were used and why were they selected for this solution?
- SQL – data extraction, joins, and querying across the five core tables
- Power BI – interactive dashboards for executive KPIs and drill-down churn analysis
- Python (scikit-learn, XGBoost) – churn-prediction modeling, selected for strong tabular-data performance
- GitHub – version control and team collaboration

Submission Assets

GitHub: [Github](#)

Email: gargisetia2006@gmail.com, tanishq2004arya@gmail.com

Dashboards, model outputs, and supporting documentation available for review upon request



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THANK YOU